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Regret Analysis of Shrinking Horizon Model Predictive Control

This paper analyzes the suboptimal implementation of shrinking horizon model predictive control (SHMPC) when a fixed number of solver iterations and a warm-start are utilized at each time-step to solve the underlying optimal control problem (OCP). We derive bounds on the loss of performance (regret) and on the difference between suboptimal SHMPC and optimal solutions. This analysis provides insights and practical guidelines for the implementation of SHMPC under computational limitations. A numerical example of axisymmetric spacecraft spin stabilization is reported. The suboptimal implementation of SHMPC is shown to be capable of steering the system from an initial state into a known terminal set while satisfying control constraints. [DOI: 10.1115/1.4066317]

Keywords: model predictive control, constrained control, regret analysis, numerical methods for optimal control

1 Introduction

In shrinking horizon MPC (SHMPC), the underlying optimal control problem (OCP) is solved from the current time to a fixed end time. Recomputing the optimal solution provides feedback that can compensate for disturbances and uncertainties. However, as onboard computing power can be limited, it is of interest to characterize the properties of SHMPC when only a finite number of optimization iterations are performed while using warm-starting.

Although SHMPC is investigated less than the more common receding horizon variant [1], its utility has already been successfully verified in some applications, such as helicopter ship landing and touchdown maneuvers [2], energy management [3], and safe navigation procedures [4]. In all of these settings, solving an online optimization problem can be computationally expensive, and this motivates research into suboptimal online algorithms as well as quantification of the impact of suboptimality on closed-loop performance.

In Ref. [5], control sequences and cost functions obtained from a receding horizon LQ-MPC and a suboptimal variant of the same MPC are compared and the incurred suboptimality is quantified. This analysis is extended to nonlinear time-varying systems in Ref. [6]. The analysis in Refs. [5] and [6] addresses the receding

horizon rather than the shrinking horizon setting. Furthermore, as opposed to the previously cited papers, in our work, we also consider the impact of disturbances and terminal constraints.

References [7–9] derive bounds on state and control trajectory deviations of SHMPC versus a continuous time optimal control; the bounds depend on the discretization time-step, state measurement accuracy, and disturbance preview/model parameter accuracy. These references do not consider the impact of inexact computations of the solution or the loss of performance in terms of cost (regret). Reference [10] considered the case of discrete-time linear systems with a disturbance input. The paper analyzes suboptimal SHMPC for the case when the cost function includes a terminal cost and there are input constraints. The current state is assumed to be accurately measured/estimated and the OCP for the model with zero disturbance is solved suboptimally with a solver executing a finite number of iterations and utilizing warm-starting. Theoretical bounds on the minimum number of iterations are derived which ensure that the closed-loop trajectory with SHMPC is in terminal set at the end of the horizon.

In this paper, we extend the analysis in Ref. [10] to regret analysis. More specifically, we derive bounds on the loss of performance (in terms of cost function deviation) due to suboptimal SHMPC implementation with a finite number of solver iterations and warm-starting. These bounds provide insight and practical guidelines for formulating the OCP in SHMPC. To illustrate the use of suboptimal SHMPC and compare the analytically expected and empirically observed trends, we consider a case study involving a problem of axisymmetric spacecraft spin stabilization.

Notation: Let $\mathbb{R}$ be the set of real numbers and $\mathbb{N}$ represent the set of positive integers including 0. Let $\mathbb{R}_{>0}$ denote the strictly positive

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real numbers (define $\mathbb{R}_{\geq 0}$). Given $a, b \in \mathbb{R}$ such that $a \leq b$ let $\mathbb{N}_{[a,b]} := \mathbb{N} \cap [a, b]$ and $\mathbb{R}_{\geq a} := \mathbb{R} \cap [a, \infty)$ (similarly for $\mathbb{R}_{>a}$). Given $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$, let $(x, y) = [x^T y^T]^T$. Let $\langle \cdot, \cdot \rangle$ denote the Euclidean inner-product. Let $\mathbb{S}_{++}^n$ and $\mathbb{S}_+^n$ denote the set of positive definite and positive semidefinite symmetric matrices. Let $\|\cdot\|$ represent the two-norm. Given $x \in \mathbb{R}^n$ and $W \in \mathbb{S}_{++}^n$, the W -norm is $\|x\|_W = \sqrt{x^T W x}$. Given two matrices $A > 0$ and $B > 0$, let $\lambda_{\min}^B(A)$ and $\lambda_{\max}^B(A)$ denote the minimum and maximum eigenvalues of $B^{-1}AB^{-1}$ which satisfy $\lambda_{\min}^B(A)\|x\|_B^2 \leq \|x\|_A^2 \leq \lambda_{\max}^B(A)\|x\|_B^2$. The identity matrix of dimension $n \times n$ is denoted by I_n while $0_{n \times m}$ denotes a zero-matrix of dimension $n \times m$. The projection of a vector x on a nonempty, closed convex set $\mathcal{A}$ is denoted by $\Pi_{\mathcal{A}}[x] := \arg\min_{y \in \mathcal{A}} \|x - y\|$. Given a set $A \subseteq \mathbb{R}^n$ and $0 < m \in \mathbb{N}$, the m times Cartesian product of A is $A^m := A \times \cdots \times A$.

2 Problem Setting

Consider a system represented by a discrete-time linear-time invariant model of the form

$$x_{k+1} = Ax_k + Bu_k + d_k \quad (1)$$

where $k \in \mathbb{N}$ denotes the time instant, $x_k \in \mathbb{R}^{n_x}$ is the state of the system, $u_k \in \mathbb{R}^{n_u}$ is the control input, $d_k \in \mathcal{D}$ is an unknown disturbance, and $\mathcal{D} \subset \mathbb{R}^{n_x}$ is a known set that bounds the disturbance input. The control objective is to steer the state of Eq. (1) from an initial state x_0 into a known terminal set Ω in $N \in \mathbb{N}$ time steps, while satisfying control constraints, so that

$$x_N \in \Omega, \quad u_k \in \mathcal{U}, \quad \forall k \in \mathbb{N}_{[0, N-1]} \quad (2)$$

where $\Omega \subset \mathbb{R}^{n_x}$ and $\mathcal{U} \subset \mathbb{R}^{n_u}$ are the terminal set and input constraint sets, respectively.

The following assumptions are made on the system under analysis.

ASSUMPTION 1. The pair (A, B) is stabilizable and the set $\mathcal{D}$ is compact and contains the origin in its interior.

ASSUMPTION 2. The set $\mathcal{U}$ is closed, convex, and contains the origin in its interior.

To address the mentioned control objectives, a SHMPC approach is used. In order to generate an input at time instant $k \in \mathbb{N}_{[0, N-1]}$, for a given $x \in \mathbb{R}^{n_x}$, the SHMPC uses the solution to a parametrized OCP, $\mathcal{P}_{N-k}(x)$, defined as

$$\min_{z_k} J_{N-k}(x, z_k) := \lambda F(\xi_N) + \sum_{i=k}^{N-1} l(\xi_i, \mu_i) \quad (3a)$$

$$\text{s.t. } \xi_k = x \quad (3b)$$

$$\xi_{i+1} = A\xi_i + B\mu_i, \quad \forall i \in \mathbb{N}_{[k, N-1]} \quad (3c)$$

$$\mu_i \in \mathcal{U}, \quad \forall i \in \mathbb{N}_{[k, N-1]} \quad (3d)$$

where $J_{N-k} : \mathbb{R}^{n_x} \times \mathbb{R}^{(N-k)n_u} \rightarrow \mathbb{R}_{\geq 0}$ is the cost function. The functions $l : \mathbb{R}^{n_x} \times \mathbb{R}^{n_u} \rightarrow \mathbb{R}_{\geq 0}$ and $F : \mathbb{R}^{n_x} \rightarrow \mathbb{R}_{\geq 0}$ are the stage and terminal costs, respectively. Moreover, in Eq. (3a), $\lambda \geq 1$ is a penalty parameter for the terminal cost. Finally, $z_k := (\mu_k, \dots, \mu_{N-1})$.

Remark 1. The OCP defined in Eq. (3) does not explicitly include the terminal set constraint introduced in Eq. (2). Instead, and following Ref. [11], a penalty parameter $\lambda \geq 1$ is allocated to the terminal cost. As shown in Ref. [11], if λ is sufficiently large, the terminal constraint Eq. (2) is satisfied. More specifically, for all elements in the N steps backward reachable set to the interior of the terminal constraint considering input constraints, there exists a minimum value of λ such that for any larger value of λ , the solution of the OCP satisfies the terminal constraint implicitly. The use of the penalty parameter avoids the need to explicitly impose the terminal

state constraints, simplifying both the optimization problem to be solved online and the optimality analysis. Section 3 provides a more detailed discussion on the relationship between λ and the terminal set.

In this work, we focus on quadratic stage and terminal cost functions that can be expressed as $l(x, u) := \|x\|_Q^2 + \|u\|_R^2$ and $F(x) := \|x\|_P^2$. Furthermore, we make the following assumption on Q , R , and P .

ASSUMPTION 3. The matrices $Q \in \mathbb{S}_{++}^{n_x}$, $R \in \mathbb{S}_{++}^{n_u}$, and $P \in \mathbb{S}_{++}^{n_x}$ are the solution to the discrete algebraic Riccati equation, $P = Q + K^T R K + (A - BK)^T P (A - BK)$, with $K = (R + B^T P B)^{-1} (B^T P A)$ the associated linear quadratic regulator gain.

For quadratic stage and terminal cost, $\mathcal{P}_{N-k}(x)$ can be reformulated in compact form as

$$\min_{z_k \in \mathcal{Z}_k} \|(x, z_k)\|_{M_k}^2 \quad (4)$$

where $\mathcal{Z}_k = \mathcal{U}^{N-k}$, $M_k = \begin{bmatrix} W_k & G_k^T \\ G_k & H_k \end{bmatrix}$, and an expression of matrices W_k , G_k , H_k can be found in Ref. [12].

For any $k \in \mathbb{N}_{[0, N-1]}$, the optimal solution mapping of Eq. (4), mapping from $x \in \mathbb{R}^{n_x}$ to the minimizer of $\mathcal{P}_{N-k}(x)$, is then given by

$$z_k^*(x) := \arg \min_{z_k \in \mathcal{Z}_k} J_{N-k}(x, z_k) \quad (5)$$

Note that, the solution mapping Eq. (5) is a function by convexity of Eq. (4). Additionally, for each $k \in \mathbb{N}_{[0, N-1]}$, we define the value function $V_{N-k} : \mathbb{R}^{n_x} \rightarrow \mathbb{R}_{\geq 0}$ associated with $\mathcal{P}_{N-k}$ as

$$V_{N-k}(x) = J_{N-k}(x, z_k^*(x)) := \min_{z_k \in \mathcal{Z}_k} J_{N-k}(x, z_k) \quad (6)$$

Moreover, the optimal solution mapping and the value function are Lipschitz continuous, see, e.g., Ref. [12]. More precisely, for the optimal solution mapping, the following result from Ref. [12] holds.

LEMMA 1. Let Assumptions 1–3 hold, it then follows that for all $k \in \mathbb{N}_{[0, N-1]}$ and $x, y \in \mathbb{R}^{n_x}$, the solution mapping defined in Eq. (5) satisfies:

$$\|z_k^*(x) - z_k^*(y)\| \leq \|H_k^{-\frac{1}{2}}\| \|G_k(x - y)\|_{H_k^{-1}} \leq L_k \|x - y\| \quad (7)$$

$$\langle z_k^*(x) - z_k^*(y), G_k(x - y) \rangle \leq -\|z_k^*(x) - z_k^*(y)\|_{H_k}^2 \quad (8)$$

where $L_k := \|H_k^{-\frac{1}{2}}\| \cdot \|H_k^{-\frac{1}{2}} G_k\|$.

Remark 2. The work [12] focuses on input-constrained RHMPC while here we consider a SHMPC formulation. Nevertheless, at each time instant, k , the OCP defined in Eq. (4) is identical to the one considered in Ref. [12] considering a horizon length $N - k$. Hence, properties such as Lipschitz continuity of the solution mapping hold, i.e., Lemma 1, holds true.

In this work, we aim to quantify the loss of performance due to the disturbance inputs entering system Eq. (1) and the use of a suboptimal SHMPC for input generation, with respect to an idealized closed-loop system. Suboptimal SHMPC approximately solves $\mathcal{P}_{N-k}(x_k)$ and selects the input from the associated vector of decision variables. Applying suboptimal SHMPC to Eq. (1) results in the following closed-loop dynamics:

$$x_{k+1} = Ax_k + Bu_k + d_k \quad (9a)$$

$$u_k = S_k z_k(x_k) := S_k(z_k^*(x_k) + e_k) \quad (9b)$$

where $z_k(x_k)$ is the approximate minimizer of $\mathcal{P}_{N-k}(x_k)$, then the SHMPC generated control input is given by $u_k = S_k z_k^*(x_k)$, where $S_k := [I_{n_u} \ 0_{n_u \times (N-k-1)n_u}]$ is a selection matrix used to select μ_k from z_k , and $e_k := z_k(x_k) - z_k^*(x_k)$ represents the error with respect to the minimizer of the OCP.

To quantify the degradation in performance due to the presence of disturbance inputs and suboptimal input generation, in this paper, we introduce the following performance loss metric defined as the cost difference over N steps for two control input sequences $\bar{u}, u^* \in \mathbb{R}^{Nn_u}$ and given the initial condition x_0 :

$$\mathcal{R}(x_0, \bar{u}, u^*) := J_N(x_0, \bar{u}) - J_N(x_0, u^*) \quad (10)$$

where $\bar{u}, u^*$ correspond to the suboptimal control sequence applied to the perturbed system and to the optimal control sequence applied to the nominal system, respectively. Then, the goal is to define a bound on the previous metric that indicates the level of approximation using the suboptimal policy. To do so, we first proceed to analyze the nominal optimal trajectory Eq. (1) and the perturbed trajectory Eq. (9). Afterward, suboptimality analysis will be explained in detail.

3 Idealized Trajectory Analysis

In this section, we review conditions under which the inexact solution of $\mathcal{P}_{N-k}(\cdot)$ results in disturbance-free trajectories reaching the terminal set defined in Eq. (2). The reader can refer to Ref. [10] for a detailed discussion of these aspects. First, we define the nominal optimal trajectory of Eq. (1) starting from $x \in \mathbb{R}^{n_x}$ at time $i \in \mathbb{N}_{[0, N-1]}$ through the recursion

$$x_{k+1}^* = Ax_k^* + Bu_k^*(x_k^*), \quad \forall k \in \mathbb{N}_{[i, N-1]} \quad (11)$$

ASSUMPTION 4. The terminal set in Eq. (2) is defined using a real number $\alpha > 0$ so that

$$\Omega := \{x \in \mathbb{R}^{n_x} | F(x) \leq \alpha\}, \text{ and} \quad (12)$$

$$\forall x \in \Omega, [\min_{u \in \mathcal{U}} F(Ax + Bu) - F(x) + l(x, u)] \leq 0 \quad (13)$$

Choosing $\alpha > 0$ such that $\Omega \subseteq \{x \in \mathbb{R}^{n_x} | -Kx \in \mathcal{U}\}$ ensures Eqs. (12) and (13) are satisfied. Moreover, as shown in Ref. [10], a set where the OCP Eq. (3) satisfies the terminal constraint in $N - k$ steps can be defined as

$$\Gamma_{N-k} := \{x \in \mathbb{R}^{n_x} | V_{N-k}(x) \leq l(x, u_k^*(x)) + (N - k - 1)\gamma + \lambda\alpha\} \quad (14)$$

and α is taken from Assumption 4, and $\gamma = \alpha \cdot \lambda_{\min}^P(Q)$.

ASSUMPTION 5. The length of the horizon N and the value of $\lambda \geq 1$ are chosen so that $x_0 \in \text{Int}\Gamma_N$, where Γ_N is defined in Eq. (14). Moreover, there exists a known $\beta > 0$ such that $V_N(x_0) \leq l(x_0, u_0^*(x_0)) + (N - 1)\gamma + \lambda\alpha - \beta$.

The following Lemma, from Ref. [10], shows how the value function is bounded when evaluated along the ideal trajectory.

LEMMA 2. Let Assumptions 1–5 hold and consider the idealized system Eq. (11). Then $\forall k \in \mathbb{N}_{[1, N-1]}$

$$V_{N-k}(x_k^*) \leq \max\{(N - k)\gamma + \lambda\alpha - \beta, \lambda\alpha\} \quad (15a)$$

$$F(x_N^*) \leq \alpha \quad (15b)$$

Remark 3. A practical way to calculate β can be summarized as follows. If we have that $V_N(x_0) < Nd + \lambda\alpha$, then $V_N(x_0) \leq Nd + \lambda\alpha - \beta$ for any $\beta < Nd + \lambda\alpha - V_N(x_0)$. In addition, we have that $Nd + \lambda\alpha - J_N(x_0, z_0) < Nd + \lambda\alpha - V_N(x_0)$. Hence, a valid choice of β can always be selected by choosing $\beta_0 = Nd + \lambda\alpha - J_N(x_0, z_0)$.

4 Perturbed Trajectory Analysis

We now review results that extend the analysis of Sec. 3 to the system Eq. (9). By defining $\omega_k = BS_k e_k + d_k$, as the disturbance

resulting from both suboptimality and unknown dynamical disturbance input, Eq. (9) can be rewritten as

$$x_{k+1} = Ax_k + Bu_k^*(x_k) + \omega_k \quad (16)$$

Due to the presence of ω_k , even starting from $x_k \in \Gamma_{N-k}$ at time instant k , the state at time N might not lie inside the terminal set. The following Theorem, from Ref. [10], shows that for sufficiently small disturbance inputs, the terminal set constraint is satisfied.

THEOREM 1. Let Assumptions 1–5 hold and consider the closed-loop system Eq. (16). Let $\{\beta_k\}_{k=0}^N$ be any sequence such that $\beta \geq \beta_0$, $\beta_k > \beta_{k+1}$, $\beta_N = 0$, and $\beta_k \in (0, (N - k)\gamma]$ for all $k \in \mathbb{N}_{[0, N-2]}$, where β is defined in Assumption 5. For $k \in \mathbb{N}_{[0, N-1]}$, define

$$\bar{V}_k := (N - k)\gamma + \lambda\alpha - \beta_k \quad (17a)$$

$$\bar{\omega}_k := \rho_k^{-1} \left(\bar{V}_{k+1}^{\frac{1}{2}} - [\bar{V}_{k+1} - (\beta_k - \beta_{k+1})]^{\frac{1}{2}} \right) \quad (17b)$$

where $\rho_k := \lambda_{\max}^{\frac{1}{2}}(W_k)$ for $k \in \mathbb{N}_{[0, N-2]}$, $\rho_{N-1} := \sqrt{\lambda \cdot \lambda_{\max}(P)}$, and $\bar{V}_N := \lambda\alpha$. If $\|\omega_k\| \leq \bar{\omega}_k$ for all $k \in \mathbb{N}_{[0, N-1]}$, then

$$V_{N-k}(x_k) \leq \bar{V}_k, \quad \forall k \in \mathbb{N}_{[1, N-1]} \quad (18a)$$

$$F(x_N) \leq \alpha \quad (18b)$$

Moreover, $x_k \in \Gamma_{N-k}$ for all $k \in \mathbb{N}_{[1, N-1]}$, $u_k \in \mathcal{U}$ for all $k \in \mathbb{N}_{[0, N-1]}$, and $x_N \in \Omega$.

The following Corollary (see Corollary 19, Ref. [10]), gives an upper-bound on the suboptimality error e_k that ensures Theorem 1 holds.

COROLLARY 1. Assume that there exist a known constant $\bar{d} > 0$ such that $\mathcal{D} \subseteq \{d \in \mathbb{R}^{n_d} | \|d\| \leq \bar{d}\}$ and $\bar{d} < \bar{\omega}_k$, for all $k \in \mathbb{N}_{[0, N-1]}$. Define $v_k := \|W^{\frac{1}{2}}B\|$ and $\bar{e}_k := (\rho_k/v_k)(\bar{\omega}_k - \bar{d})$. If, for all $k \in \mathbb{N}_{[0, N-1]}$

$$\|e_k\| \leq \bar{e} \quad (19)$$

then Eqs. (18a) and (18b) hold under the conditions of Theorem 1.

In order to find conditions such that $\|e_k\| \leq \bar{e}_k$, we now formally study e_k , entering the dynamics Eq. (16). In this work, suboptimality-induced disturbance originates from applying a finite number of iterations of an iterative solver to solve $\mathcal{P}_{N-k}(x_k)$.

4.1 Optimizer Analysis. In this paper, we proposed to solve the OCP Eq. (3) using the projected gradient method (PGM) [13]. Given an $x \in \mathbb{R}^{n_x}$ and an input vector $z_k \in \mathbb{R}^{(N-k)n_u}$, an iteration of PGM for the cost J_{N-k} is given by the operator $\mathcal{T}_k(x, z_k)$, defined as

$$\mathcal{T}_k(x, z_k) := \Pi_N[z_k - a\nabla_z J_{N-k}(x, z_k)]_{z=z_k} \quad (20)$$

where $a \in \mathbb{R}$ is a predefined step size, defined as $\alpha := 2/(\lambda_{\max}(H) + \lambda_{\min}(H))$. Then, for any $\ell \in \mathbb{N}_{[1, \infty)}$, the operator representing ℓ iterations of the PGM for the cost J_{N-k} , i.e., $\mathcal{T}_k^\ell$ is obtained by iteratively applying Eq. (20), i.e.,

$$\mathcal{T}_k^\ell(x, z_k) = \mathcal{T}_k(x, \mathcal{T}_k^{\ell-1}(x, z_k)), \quad \mathcal{T}_k^0(x, z_k) = z_k \quad (21)$$

The PGM has a linear convergence rate as detailed in the following theorem [13].

THEOREM 2. For any $x \in \mathbb{R}^n$

$$\|\mathcal{T}_k^\ell(x, z) - z_k^*(x)\| \leq \eta_k(\ell)\|z - z_k^*(x)\| \quad (22)$$

where the convergence rate, $\mathbb{R}_{\geq 0} \rightarrow (0, 1) : \ell \rightarrow \eta_k(\ell)$ is a monotonically decreasing function of ℓ such that $\eta_k(\ell) \rightarrow 0$ as $\ell \rightarrow \infty$. For the PGM, the convergence rate can be defined as

$$\eta_k(\ell) := \left(\frac{\lambda^+(H_k) - \lambda^-(H_k)}{\lambda^+(H_k) + \lambda^-(H_k)} \right)^\ell \quad (23)$$

The dynamics of the closed-loop system Eq. (16), considering the input at time instant k to be generated from ℓ_k iterations of the PGM Eq. (21) and using the warm-starting procedure, are given by

$$x_{k+1} = Ax_k + BS_k z_k(x_k) + d_k \quad (24a)$$

$$z_k(x_k) = T_k^{\ell_k}(x_k, g_k(z_{k-1})) \quad (24b)$$

where $z_{-1} \in \mathbb{R}^{(N+1)n_u}$ is equal to zero and used to initialize the warm-starting procedure. In Eq. (24b), $g_k: \mathbb{R}^{(N-(k-1))n_u} \rightarrow \mathbb{R}^{(N-k)n_u}$ is a warm-starting procedure [14] used to speed up the convergence of the PGM and defined as: $g_k(z_{k-1}) = (\mu_k, \dots, \mu_{N-1})$, for a given $z_{k-1} = (\mu_{k-1}, \dots, \mu_{N-1})$. The number of iterations of PGM directly affects the value of the suboptimality error

$$e_k = T_k^{\ell_k}(x_k, g_k(z_{k-1})) - z_k^*(x_k), \forall k \in \mathbb{N}_{[0, N-1]} \quad (25)$$

In Theorem 1, we stated that if ω_k satisfies a certain bound, then the terminal constraint is satisfied. Moreover, based on Corollary 1, Theorem 1 is satisfied if the suboptimality error satisfies certain bounds and the exogenous disturbance is bounded by a certain constant. The following Corollary from Ref. [10] defines a bound on ℓ_k that ensures that solving the OCP Eq. (3) using a finite number of PGM iterations, and despite the presence of a disturbance input entering Eq. (16), inequalities Eqs. (18a) and (18b) hold.

COROLLARY 2. Let Assumptions 1–4 hold and consider the closed-loop system Eq. (24). For all $k \in \mathbb{N}_{[0, N-1]}$, if the number of iteration $\ell_k \geq \bar{\ell}_k$, then $\|e_k\| \leq \bar{e}_k$ as defined in Eq. (19), where $\bar{e}_0 = \eta_0^{-1}(\lambda_{\min}^2(H_0)\bar{e}_0/\|G_0x_0\|_{H_0^{-1}})$, $\bar{\ell}_k = \eta_k^{-1}(\bar{e}_k/(1+\tau_k)\bar{e}_{k-1} + \sigma_k\bar{d})$, $\tau_k = \lambda_{\min}^{-\frac{1}{2}}(H_k)\|H_k^{-\frac{1}{2}}G_kBS_k\|$, $\sigma_0=0$, and $\sigma_k = \lambda_{\min}^{-\frac{1}{2}}(H_k)\|H_k^{-\frac{1}{2}}G_k\|$ for $k \geq 1$.

Remark 4. Although in this paper we focus on the PGM, it should be noted that the proposed results can be readily applied to any iterative optimization algorithm with a q -linear convergence rate (e.g., the accelerated projected gradient method [15].)

5 Suboptimality Analysis

As stated earlier, in this work, we are interested in quantifying the degradation in performance for system Eq. (1), due to the presence of unknown disturbance inputs and the use of a suboptimal SHMPC as compared to the nominal optimal system defined in Eq. (11). To quantify the degradation in performance, we introduced in Eq. (10) the metric $\mathcal{R}(x_0, \bar{u}, u^*)$. We can now define the suboptimal and optimal control input sequence as²

$$u^* := (S_0 z_0^*(x_0), \dots, S_{N-1} z_{N-1}^*(x_{N-1}^*)) \quad (26a)$$

$$\bar{u} := (S_0 z_0(x_0), \dots, S_{N-1} z_{N-1}(x_{N-1})) \quad (26b)$$

where x_k^* is given by Eq. (11) and where x_k and $z_k(x_k)$ are given by Eq. (24). At this point, we note that

$$\|\bar{u} - u^*\| \leq \sum_{k=0}^{N-1} \|S_k(z_k(x_k) - z_k^*(x_k^*))\| \leq \sum_{k=0}^{N-1} \|\Delta\mu_k\| \quad (27)$$

where $\|\Delta\mu_k\| := \|z_k(x_k) - z_k^*(x_k^*)\|$, and we used the property that $\|S_k\| \leq 1$, for all k .

In the following, we derive two bounds on the performance loss. The first is based on the analysis of the perturbed system and its

trajectories. The second one makes use of the value functions Eq. (6) and bounds on them.

5.1 Trajectory Based Analysis. To start, we give a bound on Eq. (10) as a function of the values of $\bar{u}$, u^* .

LEMMA 3. Given $x \in \mathbb{R}^{n_x}$ and $\bar{u}, u^* \in \mathbb{R}^{Nn_u}$, it follows that:

$$|\mathcal{R}(x, \bar{u}, u^*)| \leq \lambda_{\max}(H_N)\|\Delta u\|^2 + (2\|u^{*\top}\| \|H_N\| + 2\|x^\top G_N^\top\|)\|\Delta u\| \quad (28)$$

where $\Delta u = \bar{u} - u^*$.

Proof. Starting from Eq. (10), we have

$$\begin{aligned} |J_N(x, \bar{u}) - J_N(x, u^*)| &= |u_1^\top H_N \bar{u} - u^{*\top} H_N u^* + 2x^\top G_N^\top \bar{u} - 2x^\top G_N^\top u^*| \\ &\leq |\Delta u^\top H_N \Delta u| + 2|u^{*\top} H_N \Delta u| + 2|x^\top G_N^\top \Delta u| \\ &\leq \lambda_{\max}(H_N)\|\Delta u\|^2 + 2|u^{*\top} H_N \Delta u| + 2|x^\top G_N^\top \Delta u| \\ &\leq \lambda_{\max}(H_N)\|\Delta u\|^2 + (2\|u^{*\top}\| \|H_N\| + 2\|x^\top G_N^\top\|)\|\Delta u\| \end{aligned} \quad (29)$$

The first equality comes from the definition of Eq. (4), the first inequality follows from the triangle inequality and considering the definition of Δu . The second inequality comes from the positive definiteness of H_N , and the third from the submultiplicative propriety of the norm as well as the triangle inequality. ■

We now introduce a bound on the deviation in the trajectories of the perturbed and idealized systems that makes use of the following coefficients:

$$\forall k \in \mathbb{N}_{[1, N]}, \quad c_{j,k} := \begin{cases} 1, & \text{for } j = k-1 \\ \prod_{i=j+1}^{k-1} \tilde{L}_i, & \forall j \in \mathbb{N}_{[0, k-2]} \end{cases} \quad (30)$$

where $\tilde{L}_i = \|A\| + \|B\|L_i$ and L_i are defined in Lemma 1.

LEMMA 4. Let Assumptions 1–3 hold, and consider the state trajectories Eqs. (11) and (16) both starting from x_0 . Assume there exists $\{\bar{\omega}_k\}_{k=0}^{N-1}$ such that $\|\omega_k\| \leq \bar{\omega}_k, \forall k \in \mathbb{N}_{[0, N-1]}$. It then follows that for all $k \in \mathbb{N}_{[1, N]}$:

$$\|\Delta x_k\| := \|x_k - x_k^*\| \leq \sum_{j=0}^{k-1} c_{j,k} \bar{\omega}_j \quad (31)$$

where the coefficients $c_{j,k}$ are defined in Eq. (30).

Proof. For $k=1$, $\|\Delta x_1\| = \|\omega_0\| \leq \bar{\omega}_0$. Then, we have that

$$\begin{aligned} \|\Delta x_2\| &= \|Ax_1 + Bu_1^*(x_1) + \omega_1 - Ax_1^* - Bu_1^*(x_1^*)\| \\ &\leq \tilde{L}_1 \|\Delta x_1\| + \|\omega_1\| \leq \tilde{L}_1 \|\bar{\omega}_0\| + \|\bar{\omega}_1\| \end{aligned} \quad (32)$$

where the first inequality comes from Lemma 1, as well as the triangle inequality and the submultiplicative propriety of the norm.

Now, for $k \in \mathbb{N}_{[2, N]}$, assuming that Eq. (31) holds for $k-1$, we have

$$\begin{aligned} \|\Delta x_k\| &= \|Ax_{k-1} + Bu_{k-1}^*(x_{k-1}) + \omega_{k-1} - Ax_{k-1}^* - Bu_{k-1}^*(x_{k-1}^*)\| \\ &\leq \tilde{L}_{k-1} \|\Delta x_{k-1}\| + \bar{\omega}_{k-1} \\ &\leq \tilde{L}_{k-1} \sum_{j=0}^{k-2} c_{j,k-1} \bar{\omega}_j + \bar{\omega}_{k-1} = \sum_{j=0}^{k-1} c_{j,k} \bar{\omega}_j \end{aligned} \quad (33)$$

where the first equality comes from Eqs. (11) and (16). The first inequality comes from Lemma 1 as well as the triangle inequality

²Note that in the following, we use $u_k^* = S_k z_k^*$ for an optimal control input, $u_k = S_k z_k$ for a suboptimal control input.

and the submultiplicative property of the norm. Thus, by induction, the result holds for all $k \in \mathbb{N}_{[1,N-1]}$. ■

We can now proceed to derive a bound on the difference between the input sequences $\bar{u}$ and u^* . This bound depends on the convergence rate of the PGM algorithm, $\eta_k(\cdot)$, the number of PGM iterations, l_k , as well as bounds on the suboptimality and unknown disturbance inputs, $\bar{e}_k$ and $\bar{d}$, respectively.

THEOREM 3. *Let Assumptions 1–4 hold and consider the dynamics Eq. (24). Assume that for all $k \in \mathbb{N}_{[0,N-1]}$, there exists a known positive constant $\bar{d} > 0$ such that $\|d_k\| \leq \bar{d}$ and there exists a known sequence $\{\bar{e}_k\}$ such that $\|e_k\| \leq \bar{e}_k$. Define $\bar{\omega}_k = \|B\|\bar{e}_k + \bar{d}$, $\bar{\tau}_k = \eta_k(l_k)(1 + \tau_k)$, $\bar{\sigma}_k = \eta_k(l_k)\sigma_k$, where τ_k and σ_k are defined in Corollary 2 and let $\bar{e}_{-1} = \bar{e}_0/\bar{\tau}_0$. Moreover, define $\gamma_0 = 0$ and $\gamma_k = L_k \sum_{j=0}^{k-1} c_{j,k} \bar{\omega}_j$ for $k \in \mathbb{N}_{[1,N-1]}$. Then, for all $k \in \mathbb{N}_{[0,N-1]}$*

$$\|\Delta\mu_k\| \leq \bar{\tau}_k \bar{e}_{k-1} + \bar{\sigma}_k \bar{d} + \gamma_k \quad (34)$$

Proof. Let $k = 0$

$$\|\Delta\mu_0\| = \|\mathcal{T}_0^{\ell_0}(x_0, 0) - z_0^*(x_0)\| = \|e_0\| \leq \bar{e}_0 \quad (35)$$

where the equality follows from Theorem 2 and, the inequality from Lemma 1 and from the definition of $\bar{e}_0$. Now, for any $k \in \mathbb{N}_{[1,N-1]}$

$$\begin{aligned} \|\Delta\mu_k\| &= \|z_k(x_k) - z_k^*(x_k^*)\| \\ &= \|z_k(x_k) - z_k^*(x_k) + z_k^*(x_k) - z_k^*(x_k^*)\| \\ &\leq \|z_k(x_k) - z_k^*(x_k)\| + \|z_k^*(x_k) - z_k^*(x_k^*)\| \end{aligned} \quad (36)$$

For the first term on the right-hand side of Eq. (36), we have

$$\begin{aligned} \|z_k(x_k) - z_k^*(x_k)\| &= \|\mathcal{T}_k^{\ell_k}(x_k, g_k(z_{k-1})) - z_k^*(x_k)\| \\ &\leq \eta_k(l_k) \|g_k(z_{k-1}) - z_k^*(x_k)\| \\ &\leq \eta_k(l_k) \|g_k(z_{k-1}) - g_k(z_{k-1}^*(x_{k-1}))\| \\ &\quad + \eta_k(l_k) \|z_k^*(x_k) - g_k(z_{k-1}^*(x_{k-1}))\| \\ &\leq \eta_k(l_k) (\|e_{k-1}\| + \|z_k^*(x_k) - g_k(z_{k-1}^*(x_{k-1}))\|) \\ &= \eta_k(l_k) (\|e_{k-1}\| + \|z_k^*(x_k) - z_k^*(\tilde{x}_k)\|) \end{aligned} \quad (37)$$

where $\tilde{x}_k = Ax_{k-1} + Bu_k^*(x_{k-1})$ is the ideal state update given x_k . In Eq. (37), the first inequality is based on Theorem 2, the second inequality makes use of the triangle inequality, the third follows from the definition of the warm-starting procedure g_k , and the last equality holds from the principle of optimality for the ideal state update. We now analyze the second term of the right-hand side of Eq. (37)

$$\begin{aligned} \|z_k^*(x_k) - z_k^*(\tilde{x}_k)\| &\leq \lambda_{\min}^{-\frac{1}{2}}(H_k) \|z_k^*(x_k) - z_k^*(\tilde{x}_k)\|_{H_k} \\ &\leq \lambda_{\min}^{-\frac{1}{2}}(H_k) \|G_k(x_k - \tilde{x}_k)\|_{H_k^{-1}} \\ &\leq \lambda_{\min}^{-\frac{1}{2}}(H_k) \|H_k^{-\frac{1}{2}} G_k(d_{k-1} + BS_{k-1}e_{k-1})\| \\ &\leq \tau_k \|e_{k-1}\| + \sigma_k d_{k-1} \end{aligned} \quad (38)$$

where the second inequality comes from Lemma 1 and the third inequality from Eq. (16) and the definition of the ideal state update. Combining Eqs. (37) and (38) and considering the assumptions that $\|e_{k-1}\| \leq \bar{e}_{k-1}$ and $\|d_{k-1}\| \leq \bar{d}$, then

$$\|z_k(x_k) - z_k^*(x_k)\| \leq \bar{\tau}_k \bar{e}_{k-1} + \bar{\sigma}_k \bar{d} \quad (39)$$

We now bound the second term of the right-hand side of Eq. (36). First, we note that given $\bar{d}$, $\{\bar{e}_k\}_{k=0}^{N-1}$, and $\{\bar{\omega}_k\}_{k=0}^{N-1}$, the trajectory of Eq. (24) considered here will be equivalent to a trajectory of

Eq. (16), starting from the same initial condition and with the disturbance input respecting: $\|\omega_k\| \leq \bar{\omega}_k$. Then, using Lemma 1 and Lemma 4, we get

$$\|z_k^*(x_k) - z_k^*(x_k^*)\| \leq L_k \|x_k - x_k^*\| \leq L_k \sum_{j=0}^{k-1} c_{j,k} \bar{\omega}_j \quad (40)$$

Inserting, Eqs. (39) and (40) into Eq. (36), we obtain

$$\|\Delta\mu_k\| \leq \bar{\tau}_k \bar{e}_{k-1} + \bar{\sigma}_k \bar{d} + \gamma_k \quad (41)$$

which concludes the proof. ■

The following Theorem derives a first bound on the performance loss.

THEOREM 4. *Let Assumptions 1–4 hold, and let $\bar{u}$, u^* be defined as in Eq. (26), i.e., generated from the dynamics Eqs. (24) and (11), respectively. Assume that for all $k \in \mathbb{N}_{[0,N-1]}$, there exists a known positive constant $\bar{d} > 0$ such that $\|d_k\| \leq \bar{d}$ and there exists a known sequence $\{\bar{e}_k\}$ such that $\|e_k\| \leq \bar{e}_k$. Define $\bar{\omega}_k = \|B\|\bar{e}_k + \bar{d}$ and $\delta_k = \bar{\tau}_k \bar{e}_{k-1} + \bar{\sigma}_k \bar{d} + \gamma_k$, for all $k \in \mathbb{N}_{[1,N-1]}$, where $\bar{\tau}_k$, $\bar{\sigma}_k$, γ_k , and $\bar{e}_{-1}$ are defined in Theorem 3. Then, it follows that:*

$$\mathcal{R}(x_0, \bar{u}, u^*) \leq a \sum_{k=0}^{N-1} \delta_k^2 + 2a \sum_{k=1}^{N-1} \sum_{j=0}^{k-1} \delta_k \delta_j + \bar{b} \sum_{k=0}^{N-1} \delta_k \quad (42)$$

where $a = \lambda_{\max}(H_N)$ and $\bar{b} = 2\|x_0\|(\|L_N\|H_N\| + \|G_N^T\|)$ and $x_0 \in \mathbb{R}^{n_x}$.

Proof. First, considering the second term of the right-hand side of Eq. (28), we get

$$2\|u^{*\top}\| \|H_N\| + 2\|x^T G_N^T\| \leq \bar{b} \quad (43)$$

using the optimality of u^* and Lemma 1. Now, starting from Lemma 3, we have

$$\begin{aligned} \mathcal{R}(x_0, \bar{u}, u^*) &\leq a \|\Delta u\|^2 + \bar{b} \|\Delta u\| \\ &\leq a \left(\sum_{k=0}^{N-1} \|\Delta\mu_k\| \right)^2 + \bar{b} \sum_{k=0}^{N-1} \|\Delta\mu_k\| \\ &\leq a \left(\sum_{k=0}^{N-1} \delta_k \right)^2 + \bar{b} \sum_{k=0}^{N-1} \delta_k \\ &\leq a \sum_{k=0}^{N-1} \delta_k^2 + 2a \sum_{k=1}^{N-1} \sum_{j=0}^{k-1} \delta_k \delta_j + \bar{b} \sum_{k=0}^{N-1} \delta_k \end{aligned} \quad (44)$$

where the first inequality comes from Eq. (43) and positive definiteness of the norm, the second inequality comes from Eq. (27), and the third inequality comes from Theorem 3. ■

The bound presented above is general in the sense that it does not require a minimum number of iterations to be performed and in particular it does not require the perturbed system trajectory to enter the terminal set. However, it can be very conservative (see Numerical Examples, Sec. 6), mainly due to the various sum and products of the Lipschitz constants of the optimal value mappings, entering through the $c_{j,k}$ coefficients and appearing in δ_k . In Sec. 5.2, we propose another way to bound the performance loss based on the value function and making use of results from Ref. [10], see Sec. 4 of this paper for a summary of the relevant results.

5.2 Level-Set Based Analysis. We start with two intermediary results derived from Lemma 2 and from Theorem 1 that bound the idealized state trajectory Eq. (11) and the perturbed state trajectory Eq. (16).

LEMMA 5. *Let Assumptions 1–5 hold and consider the idealized system Eq. (11). Then $\forall k \in \mathbb{N}_{[1,N]}$*

$$\|x_k^*\| \leq \psi_k \quad (45)$$

where $\psi_N := \sqrt{\alpha}/\sqrt{\lambda_{\min}(P)}$ and, for all $k \in \mathbb{N}_{[1,N-1]}$

$$\psi_k := \frac{\sqrt{\max\{(N-k)\gamma + \lambda\alpha - \beta, \lambda\alpha\}}}{\sqrt{\lambda_{\min}(Q)}} \quad (46)$$

Proof. For all $k \in \mathbb{N}_{[1,N-1]}$

$$\|x_k^*\|_Q^2 \leq V_{N-k}(x_k^*) \leq \max\{(N-k)\gamma + \lambda\alpha - \beta, \lambda\alpha\} \quad (47)$$

where the first inequality comes from Eq. (3a) considering the quadratic stage and terminal cost functions and the second inequality from Lemma 2. Then, given that $\sqrt{\lambda_{\min}(Q)}\|x_k^*\| \leq \|x_k^*\|_Q$ we get $\|x_k^*\| \leq \psi_k$. The same arguments can be applied to $\|x_N^*\|_P$ to get $\|x_N^*\| \leq \psi_N$. ■

LEMMA 6. Let Assumptions 1–5 hold and let $\{\beta_k\}_{k=0}^N$ and $\{\bar{\omega}_k\}_{k=0}^{N-1}$ be defined as in Theorem 1. Consider the perturbed state trajectory Eq. (16), if, for all $k \in \mathbb{N}_{[0,N-1]}$, $\|\omega_k\| \leq \bar{\omega}_k$, then, for all $k \in \mathbb{N}_{[1,N]}$

$$\|x_k\| \leq \bar{\psi}_k \quad (48)$$

where $\bar{\psi}_N := \sqrt{\alpha}/\sqrt{\lambda_{\min}(P)}$ and, for all $k \in \mathbb{N}_{[1,N-1]}$

$$\bar{\psi}_k := \frac{\sqrt{(N-k)\gamma + \lambda\alpha - \beta_k}}{\sqrt{\lambda_{\min}(Q)}} \quad (49)$$

Proof. The proof follows that of Lemma 5 by deriving bounds on the value function evaluated along the perturbed trajectory given in Theorem 1. ■

Based on the results above, the following result provides an alternative to Theorem 3 for bounding $\|\Delta\mu_k\|$.

COROLLARY 3. Let Assumptions 1–5 hold, consider the dynamics Eqs. (11) and (24), and assume that $\{\bar{\omega}_k\}_{k=0}^{N-1}$ and $\{\beta_k\}_{k=0}^N$ are defined as in Theorem 1. Moreover, assume that the bounds on the minimum number of iterations defined in Corollary 2 are respected so that Eq. (19) holds and define $\{\bar{e}_k\}_{k=0}^{N-1}$ accordingly. For all $k \in \mathbb{N}_{[0,N-1]}$, define $\bar{\tau}_k = \eta_k(\ell_k)(1 + \tau_k)$, $\bar{\sigma}_k = \eta_k(\ell_k)\sigma_k$, where τ_k and σ_k are defined in Corollary 2 and let $\bar{e}_{-1} = \bar{e}_0/\bar{\tau}_0$. Moreover, let $\bar{\gamma}_0 = 0$ and, for $k \in \mathbb{N}_{[1,k-1]}$, $\bar{\gamma}_k = L_k(\bar{\psi}_k + \psi_k)$. Then, for all $k \in \mathbb{N}_{[0,N-1]}$

$$\|\Delta\mu_k\| \leq \bar{\tau}_k \bar{e}_{k-1} + \bar{\sigma}_k \bar{d} + \bar{\gamma}_k \quad (50)$$

where $\|\Delta\mu_k\| := \|z_k(x_k) - z_k^*(x_k^*)\|$.

Proof. The proof follows that of Theorem 3 up to the derivation of the bound on the second term of the right-hand of Eq. (36). Instead of using Eq. (40), the second term of the right-hand of Eq. (36) can be bounded as following:

$$\begin{aligned} \|z_k^*(x_k) - z_k^*(x_k^*)\| &\leq L_k \|x_k - x_k^*\| \\ &\leq L_k (\|x_k\| + \|x_k^*\|) \leq L_k (\bar{\psi}_k + \psi_k) \end{aligned} \quad (51)$$

where we used Lemma 1, the fact that Lemma 5 and Lemma 6 can be applied to the trajectories considered here, using the assumption on the minimum number of iterations of PGM. ■

Before we present a second bound on the performance loss, we introduce the coefficients

$$h_0 = 1, h_k = \prod_{i=0}^{k-1} \|A\| + \|B\|L_i, \forall k \in \mathbb{N}_{[1,N]} \quad (52)$$

where the L_i are defined in Lemma 1.

THEOREM 5. Let Assumptions 1–4 hold, and let $\bar{u}$, u^* be defined as in Eq. (26), i.e., generated from the dynamics Eqs. (24) and (11),

respectively. Assume that $\{\bar{\omega}_k\}_{k=0}^{N-1}$ and $\{\beta_k\}_{k=0}^N$ are defined as in Theorem 1. Moreover, assume that the bounds on the minimum number of iterations defined in Corollary 2 are respected so that Eq. (19) holds and define $\{\bar{e}_k\}_{k=0}^{N-1}$ accordingly. Define $\delta_k = \bar{\tau}_k \bar{e}_{k-1} + \bar{\sigma}_k \bar{d} + \bar{\gamma}_k$, for all $k \in \mathbb{N}_{[1,N-1]}$, where $\bar{\tau}_k$, $\bar{\sigma}_k$, $\bar{\gamma}_k$, and $\bar{e}_{-1}$ are defined in Corollary 3. Then, it follows that:

$$\mathcal{R}(x_0, \bar{u}, u^*) \leq \bar{q} \sum_{k=0}^N \bar{\psi}_k^2 + \sum_{k=0}^{N-1} \|R\| (\delta_k + 2h_k \psi_k) \delta_k \quad (53)$$

where $\bar{q} = \max\{\|Q\|, \lambda\|P\|\}$ and the coefficients h_k are given in Eq. (52).

Proof. Starting from the definition of $J_N(x, \cdot)$, we have

$$\begin{aligned} \mathcal{R}(x_0, \bar{u}, u^*) &= \lambda (\|x_N\|_P^2 - \|x_N^*\|_P^2) \\ &\quad + \sum_{k=0}^{N-1} (\|x_k\|_Q^2 - \|x_k^*\|_Q^2) + \sum_{k=0}^{N-1} \|\bar{u}_k\|_R^2 - \|u_k^*\|_R^2 \end{aligned} \quad (54)$$

First, we focus on the input terms

$$\begin{aligned} &\sum_{k=0}^{N-1} \|\bar{u}_k\|_R^2 - \|u_k^*\|_R^2 \\ &= \sum_{k=0}^{N-1} \|S_k z_k(x_k)\|_R^2 - \|S_k z_k^*(x_k^*)\|_R^2 \\ &= \sum_{k=0}^{N-1} \|S_k (\Delta\mu_k + z_k^*(x_k^*))\|_R^2 - \|S_k z_k^*(x_k^*)\|_R^2 \\ &\leq \sum_{k=0}^{N-1} \|S_k \Delta\mu_k\|_R^2 + 2\|S_k \Delta\mu_k\|_R \|S_k z_k^*(x_k^*)\|_R \\ &\leq \|R\| \sum_{k=0}^{N-1} \|\Delta\mu_k\|^2 + 2\|\Delta\mu_k\| \|z_k^*(x_k^*)\| \\ &\leq \|R\| \sum_{k=0}^{N-1} \delta_k^2 + 2\delta_k \|z_k^*(x_k^*)\| \leq \|R\| \sum_{k=0}^{N-1} \delta_k^2 + 2L_k \delta_k \|x_k^*\| \\ &\leq \|R\| \sum_{k=0}^{N-1} (\delta_k + 2L_k \psi_k) \delta_k \end{aligned} \quad (55)$$

where the first equality comes from Eq. (26) and the second from the definition of $\Delta\mu_k$. Then, the first inequality follows from the Cauchy–Schwarz inequality. The second inequality comes from $\|u\|_R^2 \leq \lambda_{\max}(R)\|u\|^2 = \|R\|\|u\|^2$ for all $u \in \mathbb{R}^{n_u}$ as $R \in \mathbb{S}_{++}^{n_u}$, the submultiplicative property of the norm and $\|S_k\| \leq 1$ for all k . The third inequality comes from Corollary 3, and the fourth from Lemma 1. The last inequality follows from Lemma 5.

Regarding the state terms

$$\begin{aligned} &\lambda (\|x_N\|_P^2 - \|x_N^*\|_P^2) + \sum_{k=0}^{N-1} (\|x_k\|_Q^2 - \|x_k^*\|_Q^2) \\ &\leq \bar{q} \sum_{k=0}^N (\|x_k\|^2 - \|x_k^*\|^2) \leq \bar{q} \sum_{k=0}^N \bar{\psi}_k^2 \end{aligned} \quad (56)$$

where the last inequality comes from Lemma 5 and Lemma 6 and considering that $\|x_k^*\|^2 \geq 0$ for all $k \in \mathbb{N}_{[0,N]}$. Combining Eqs. (54) and (56), we get the desired result. ■

6 Numerical Examples

In this section, we illustrate the result obtained in Sec. 5 by considering a problem of axisymmetric spacecraft spin stabilization

[16]. The continuous-time equations of motion for the transversal components θ_1 and θ_2 of the angular velocity vector are

$$\ddot{\theta}_1 = n \left(\frac{I_t - I_3}{I_t} \right) \dot{\theta}_2 + \frac{M_d}{I_t} \quad (57)$$

$$\ddot{\theta}_2 = -n \left(\frac{I_t - I_3}{I_t} \right) \dot{\theta}_1 + \frac{M_c}{I_t} \quad (58)$$

where I_t is the spacecraft transversal moment of inertia, I_3 is the spacecraft moment of inertia about the spin axis, n is the spin rate, M_d is a disturbance torque, and M_c is the control moment which is subject to saturation constraints, $M_c \in \mathcal{U} = [-0.2, 0.2]$. Moreover, during the simulations, we use the following numerical values: $I_t = 1$, $I_3 = 0.05$, and $n = 1$.

The continuous-time model is discretized using a sampling period of $T = 0.1$ s, and the initial horizon length for the OCP is chosen as $N = 150$. The terminal set Ω in Assumption 4 is chosen by specifying $\alpha > 0$ as the largest value such that $\Omega \in \{x | -Kx \in \mathcal{U}\}$. The initial condition x_0 for the system is $x_0 = (1, 1)$. Moreover, the terminal penalty weight is defined as $\lambda = 10$ and the cost matrices as $Q = \text{diag}(1, 1)$ and $R = 20$. The nominal values of some of these parameters will be varied during the simulations to show their influence on the suboptimality bound [17–21].

In each of the simulation scenarios described in this section, the goal is to guide the system to a terminal set while reacting to an external disturbance. The performance of the suboptimal shrinking horizon MPC applied to the “real” system subject to a disturbance torque M_d is compared with that of an optimal shrinking horizon MPC applied to the idealized system without disturbance.

In the following, the suboptimality metric analyzed is the one reported in Theorem 5, Sec. 5.2. As explained previously, the bound presented in Theorem 4 tends to be conservative. As an illustration of this, using the nominal simulation parameters given above, the bound defined in Theorem 4 turns out to be $\mathcal{R} = 246 \times 10^{21}$, while the one defined in Theorem 5 is $\mathcal{R} = 337 \times 10^4$, where the value of suboptimal and optimal cost functions are $\bar{J} = 267.3$ and $J^* = 149.87$, respectively. The state and control trajectories are shown in Figs. 1 and 2. We used a black solid line for the optimal SHMPC and a line with circle markers for the suboptimal SHMPC.

Next, we analyze the behavior of the suboptimality bound by reducing the number of iterations required to enter the terminal set. In the following, we consider three variants of the suboptimal SHMPCs referred to as SHMPC-1, SHMPC-2, and SHMPC-3, which differ by the number of iterations from the highest to lowest. The number of iterations is determined using Corollary 2. The state and control trajectories are shown in Figs. 1 and 2. The number of iterations performed and the behavior of the bound are shown in Fig. 3. As can be seen from Figs. 1 and 2, in all examples the system is controlled so that it enters the terminal set while satisfying the

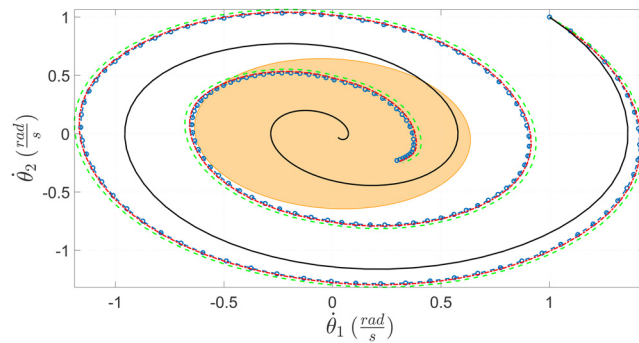


Fig. 1 State trajectories. SHMPC (solid line), SHMPC-1 (line with circle markers), SHMPC-2 (dotted line), and SHMPC-3 (dashed line). Terminal set is indicated by a round zone.

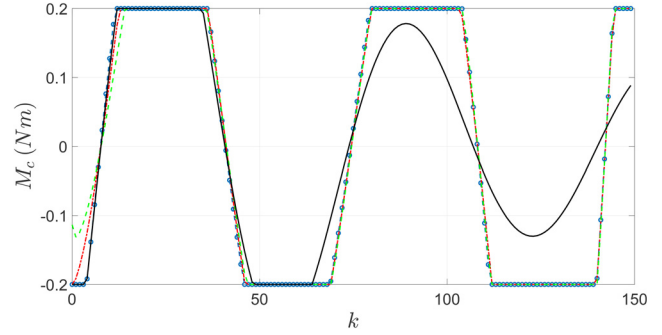


Fig. 2 Control input. Solid line optimal SHMPC. Line with circle markers suboptimal SHMPC-1. Dotted line SHMPC-2. Dashed line suboptimal SHMPC-3.

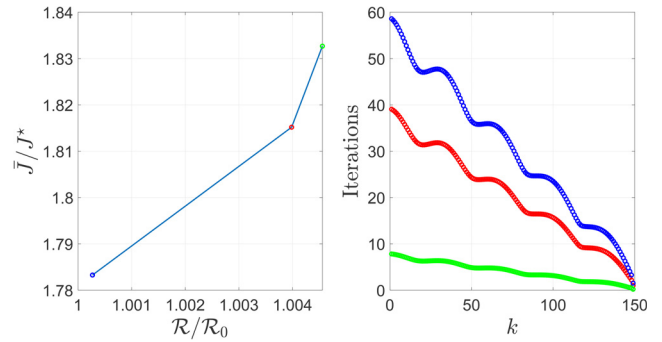


Fig. 3 Left: trend of cost function ($\bar{J}$ suboptimal cost function and J^* optimal cost function) and suboptimal bound $\mathcal{R}$ reported in Theorem 5, $\mathcal{R}_0$ nominal bound value. Right: iterations performed. Upper line: suboptimal SHMPC-1. Middle line SHMPC-2. Lower line suboptimal SHMPC-3.

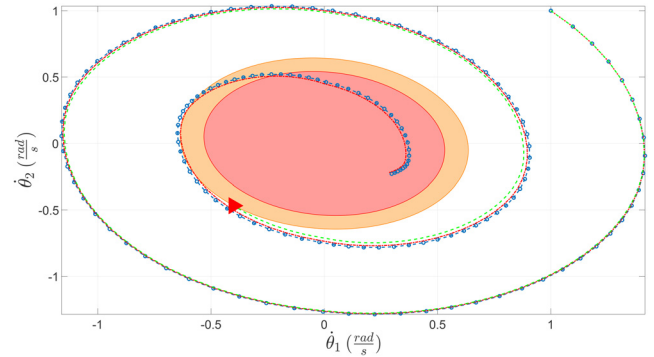


Fig. 4 State trajectories and terminal sets. Scenario 1: $N=150/R=20$. Line with circle markers suboptimal SHMPC. Outer zone: terminal set. Scenario 2: $N=150/R=15$. Dotted line SHMPC. Inner zone: terminal set. Scenario 3: $N=100/R=15$. Dashed line suboptimal SHMPC. Outer zone: terminal set. Triangles marker: end of state trajectory.

Table 1 Simulation results

	$N = 150/R = 20$	$N = 150/R = 15$	$N = 100/R = 20$
J^*	149.87	137.42	171.33
$\bar{J}$	267.3	230.83	173.05
$\mathcal{R} \times 10^4$	337	238	218

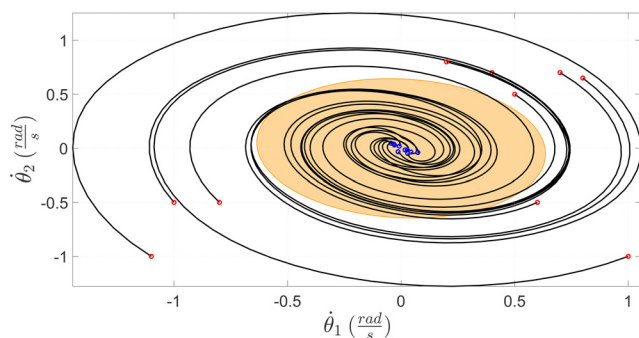


Fig. 5 State trajectories for different initial conditions and disturbance. External zoon circle marker: initial condition. Internal circle marker: final condition. Terminal set is indicated by round zone.

constraints on the control input. Note that the difference between the performance of suboptimal and optimal control schemes is due to the presence of external disturbance. Figure 3 shows the suboptimality metric and the value of the cost functions. To analyze their behavior, we considered as the nominal value $\mathcal{R}_0$ the one obtained by performing a very large number of iterations. Consistently with the expectations, the decrease in the bound value is correlated with the increase in the number of iterations performed.

We proceed to show how the suboptimality metric $\mathcal{R}$ depends on some of the control parameters. Considering Eq. (51), the values of ψ and $\tilde{\psi}$ defined in Lemmas 5 and 6 are strictly dependent on the size of the prediction horizon N and the size of the terminal set α , where α depends on how the R matrix is tuned. For this reason, we proceed to evaluate the sensitivity of the bound on $\mathcal{R}$ with respect to the horizon length N and to the size of the terminal set (i.e., the value of α). Figure 4 and Table 1 report the results of these numerical examples. By reducing the horizon length N , or by α , the bound $\mathcal{R}$ is reduced indicating that the behavior of the suboptimal control is approaching optimal. This analysis suggests the possibility of tuning the control parameters to have a bound that more accurately reflects suboptimal behavior. Note that, as shown in Fig. 4, while reducing N leads to a less conservative bound, too short a prediction horizon could jeopardize system entry into the terminal set. To conclude this section, as shown in Fig. 5, if the initial conditions is such that $x_0 \in \text{Int}\Gamma_N$, and the disturbance is such that it satisfies Corollary 1, the proposed control scheme is able to drive the system toward the terminal set.

Remark 5. The major source of conservativeness in the bound presented in Theorem 5 is due to Eq. (56). That bound comes from Lemma 6 which strictly depends on some parameters related to the specific control scheme presented in our work. The development of alternative bounds will be considered in future studies.

7 Conclusions

This paper focuses on suboptimal SHMPC, where only a fixed number of solver iterations can be performed. Through a regret analysis, we highlighted the tradeoffs between performance degradation and computational limitations. Future research directions will include refining the quality of the bounds on the regret metric, the use of uncertain disturbance preview in the OCP, and imperfect observation of the state which can be interpreted as an additive disturbance.

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Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

- I_t = transversal moment of inertia (kg m²)
- I_3 = moment of inertia about the spin axis (kg m²)
- M_c = control torque (N·m)
- M_d = disturbance torque (N·m)
- $\dot{\theta}_{1,2}$ = angular velocity (rad s⁻²)
- LQ-MPC = linear quadratic model predictive control
- MPC = model predictive control
- OCP = optimal control problem
- SHMPC = shrinking horizon model predictive control

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